

Varun Sathya

Undergraduate in Electrical and Electronics Engineering

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EDUCATION

College of Engineering Trivandrum

B.Tech Electrical & Electronics Engineering — GPA: 8.88/10

Relevant Coursework: Power Electronics, Linear Control Systems, Digital Electronics, Analog Electronics, Microprocessors and Microcontrollers, Signals & Systems

SKILLS

Tools & Software: MATLAB/Simulink, Embedded C, Arduino IDE, KiCad, Proteus, STM32, Python, C/C++, Verilog (Quartus/ModelSim)

Technical Areas: PID & Feedback Control Systems, DC-DC Converters, Motor Control (PWM, Speed/Torque Regulation), Power Electronics Design, MPPT, Embedded Systems, Heuristic Optimization (GA, PSO, Simulated Annealing)

Languages: English, Tamil, Hindi, Malayalam

EXPERIENCE

Indian Institute of Space Science and Technology

May 2026 – Present

Research Intern — Modular Electrical Power Distribution for Small Satellites

- Conducting literature survey and simulation of modular EPS architectures for small satellites (Project AV34, under Dr. Anindya Dasgupta).
- Analysing power distribution topologies and their fault tolerance, efficiency, and scalability trade-offs.
- Developing PCB layouts in KiCad for modular power distribution units.

Dept. of Electrical Engineering, College of Engineering Trivandrum

April 2026 – Present

Bridge Course Mentor — DC Motor Speed Control

- Mentored junior EEE students on PWM-based DC motor speed control using Arduino.
- Taught C/C++ programming for implementing motor control algorithms.
- Conducted real-time performance evaluation and waveform analysis using Digital Storage Oscilloscope (DSO).

Indian Institute of Space Science and Technology

June 2025

Summer Research Intern

- Developed agent-based simulations in NetLogo for [Computational Models of Complex Societies](#).
- Gained expertise in heuristic optimization by implementing Genetic Algorithms and Particle Swarm Optimization.

PROJECTS

PID-Controlled DC-DC Boost Converter with Genetic Algorithm Tuning

April 2026

Course Project — Dept. of Electrical Engineering, CET

- Designed and simulated closed-loop PID boost converter in MATLAB/Simulink (12V to regulated 20V output).
- Implemented Genetic Algorithm for optimal PID tuning to minimize steady-state error and overshoot.
- Developing 4-layer KiCad PCB and STM32 firmware in C for 25kHz PWM generation controlled by GA.

Solar PV MPPT with DC-DC Converter

Mar 2026 – Present

Project Coordinator — IET On-Campus CET

- Leading a team to develop a solar Maximum Power Point Tracking energy harvesting system.
- Designing DC-DC converter hardware and control algorithms in MATLAB and Python for efficient power extraction.

Deep Learning for Autonomous Agricultural Drone

Jan 2025

Delta Automations — State Government Funded Project

- Contributed to development of autonomous agricultural drone for weed detection and removal.
- Built semantic segmentation pipeline using DeepLabv3 architecture.
- Trained model on large image dataset for pixel-level healthy vs weed-infected region classification.

VLSI Macro Floorplanning Tool (LAIOpt)

Dec 2025 – Jan 2026

1st Place — TCS Decode Hackathon & Winner — VIA CampusConnect

- Developed hybrid Reinforcement Learning + Simulated Annealing optimizer for macro placement.
- Minimized wirelength and thermal dissipation.
- Deployed the tool as a Streamlit web application.

ACTIVITIES

- Researched and benchmarked Fuzzy Logic, Genetic Algorithm, and PSO for automated PID parameter tuning in MATLAB/Simulink.
- Built embedded control systems including Arduino-based self-balancing robot (gyroscope + PID), line follower robot, obstacle avoidance robot etc. as a part of IEEE SB CET and RoboCET.
- Prototyped geofencing safety device using Arduino Nano, GPS, and GSM modules with Proteus simulation, submitted to project expo at NIT Calicut (Tathva 2024)
- Completed NPTEL courses in Hardware Modelling using Verilog, VLSI Physical Design, and Analog Electronics.